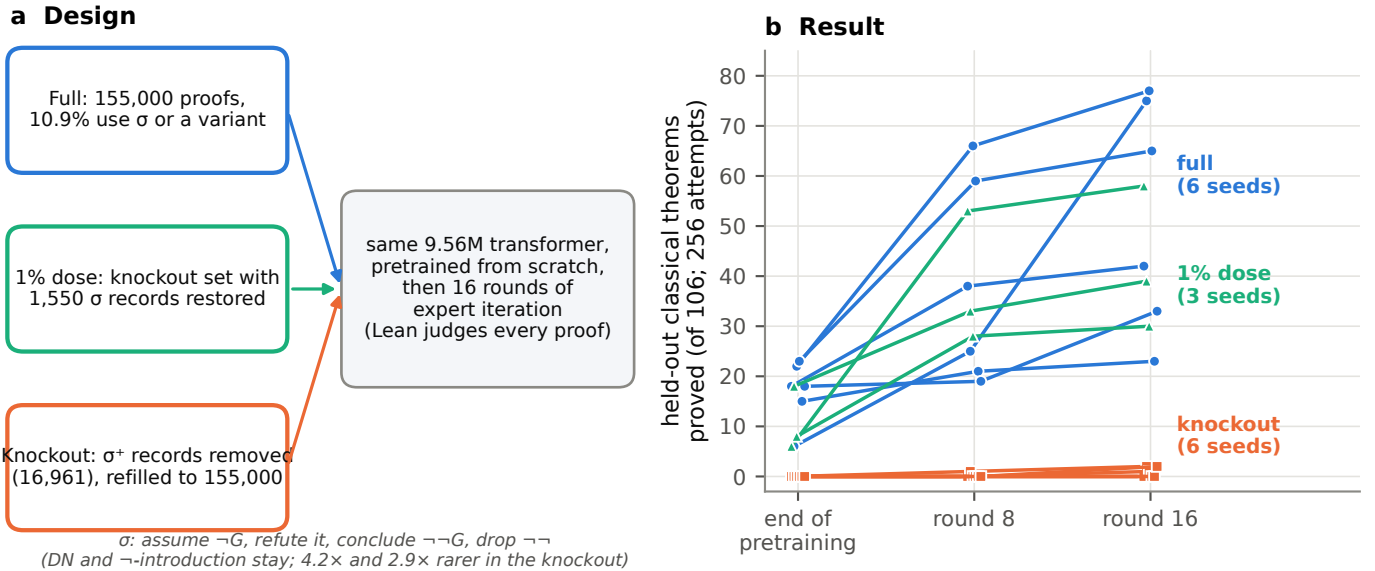
a Design
Full: 155,000 proofs, 10.9% use σ or a variant
1% dose: knockout set with 1,550 σ records restored
Knockout: σ⁺ records removed (16,961), refilled to 155,000
same 9.56M transformer, pretrained from scratch, then 16 rounds of expert iteration (Lean judges every proof)
σ: assume ¬G, refute it, conclude ¬¬G, drop ¬¬
(DN and ¬-introduction stay; 4.2× and 2.9× rarer in the knockout)
b Result
held-out classical theorems proved (of 106; 256 attempts)
0
10
20
30
40
50
60
70
80
end of pretraining
round 8
round 16
full (6 seeds)
1% dose (3 seeds)
knockout (6 seeds)